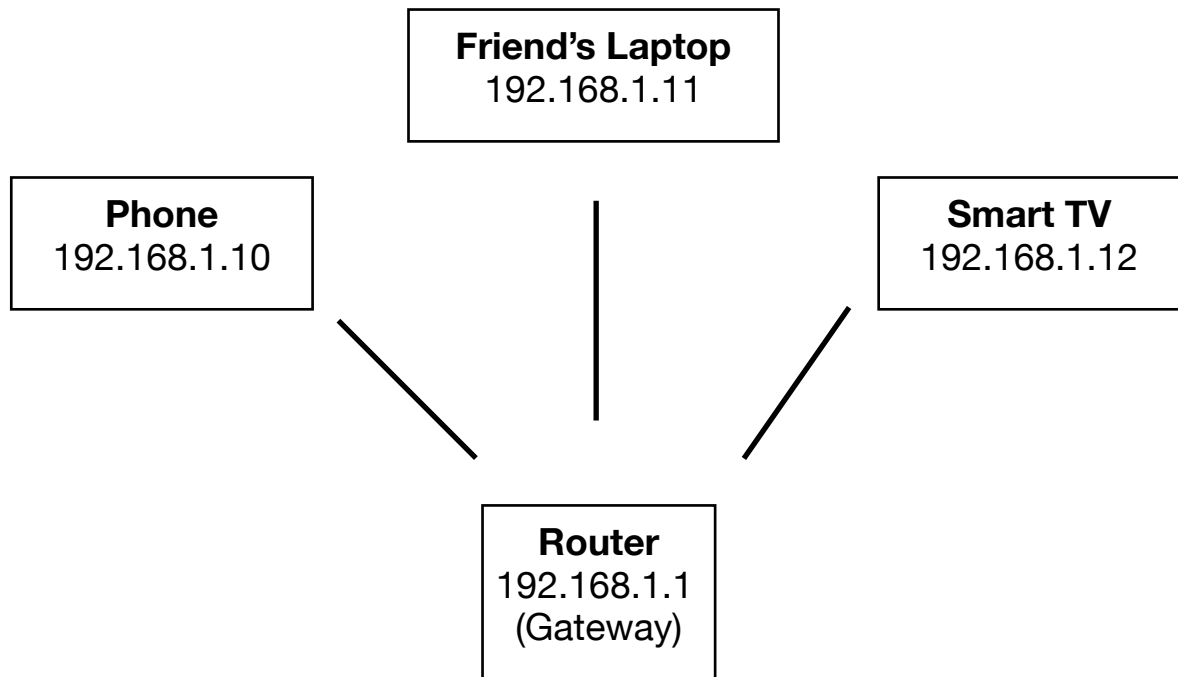


Module 1: Network + Network Fundamentals and building networks

IP Routing and Routing Protocols

Task 1:



Task 2:

Manual IP routing simulation: laptop → friend's computer (different subnet)

Setup (made-up addresses):

- Your laptop: 192.168.1.10/24, gateway 192.168.1.1
- Your friend's computer: 192.168.2.20/24, gateway 192.168.2.1
- Router R1 connects subnet 192.168.1.0/24 (interface eth0) and links to Router R2
- Router R2 connects to subnet 192.168.2.0/24 (interface eth1) where your friend's PC sits

Step-by-step packet journey:

1. Your laptop wants to send a packet to 192.168.2.20. It checks its own subnet (192.168.1.0/24) — the destination isn't in it, so it can't deliver directly.
2. Your laptop consults its routing table, finds no specific route, and sends the packet to its **default gateway**, 192.168.1.1 (Router R1).
3. **Router R1** receives the packet and checks its routing table:

Destination	Subnet mask	Next Hop	Interface
192.168.1.0	255.255.255.0	directly connected	eth0
192.168.2.0	255.255.255.0	10.0.0.2 (R2)	eth1

It matches 192.168.2.0/24 and forwards the packet out its eth1 interface toward Router R2 (10.0.0.2).

4. **Router R2** receives the packet and checks its own table:

Destination	Subnet mask	Next Hop	Interface
192.168.2.0	255.255.255.0	directly connected	Eth1

It sees 192.168.2.0/24 is directly connected on its eth1 interface, so it delivers the packet directly to your friend's computer at 192.168.2.20.

5. Each router made its decision using longest-prefix match against its routing table — no single device knew the whole path in advance; each hop just forwarded the packet one step closer based on its own local table.

Task 3:

City	Network Address	Subnet mask	Next hop	Interface
Mumbai	10.1.0.0	255.255.255.0	Directly connected	eth0
Delhi	10.2.0.0	255.255.255.0	172.16.0.2	Eth1
Bengaluru	10.3.0.0	255.255.255.0	172.16.0.3	eth2
Default route (Internet)	0.0.0.0	0.0.0.0	ISP gateway	Eth3

Task 4:

Static vs. dynamic routing

Static routing:

Pros

- 1) Predictable and secure — no risk of unauthorized route advertisements.
- 2) No extra CPU/bandwidth overhead for route calculation.
- 3) Simple to configure for small, stable networks.

Cons:

- 1) Doesn't adapt automatically if a link fails — manual reconfiguration needed.
- 2) Doesn't scale well as the network grows (e.g., adding new Zomato branches means manually updating every router).
- 3) No automatic load balancing across multiple paths.

Dynamic routing (RIP/OSPF):

Pros

- 1) Automatically reroutes around failures — e.g., if a food delivery app's data center link goes down, OSPF can instantly find another path so orders keep flowing.
- 2) Scales easily as new branches/subnets are added — routers just learn new routes automatically.
- 3) Can load-balance traffic across multiple equal-cost paths for better performance.

Cons:

- 1) More complex to configure and troubleshoot.
- 2) Consumes extra bandwidth and CPU for periodic route updates/calculations.
- 3) Potential security risk if not properly authenticated (a rogue device could inject false routes).

Task 5:

Sample OSPF configuration

Sample AI-generated configuration for two routers

```
Router1(config)# router ospf 1
```

```
Router1(config-router)# network 192.168.1.0 0.0.0.255 area 0
```

```
Router1(config-router)# network 10.0.0.0 0.0.0.3 area 0
```

```
Router2(config)# router ospf 1
```

```
Router2(config-router)# network 192.168.2.0 0.0.0.255 area 0
```

```
Router2(config-router)# network 10.0.0.0 0.0.0.3 area 0
```

- **router ospf 1** — Enables the OSPF routing process on this router; "1" is just a locally-significant process ID, not shared between routers.
- **network 192.168.1.0 0.0.0.255 area 0** — Tells OSPF "this router's directly-connected 192.168.1.0/24 subnet should participate in OSPF," using a wildcard mask (inverse of a subnet mask) instead of a normal subnet mask; area 0 places it in OSPF's backbone area.
- **network 10.0.0.0 0.0.0.3 area 0** — Enables OSPF on the small point-to-point link (a /30, only 4 addresses) connecting Router1 and Router2 directly, so they can discover each other and exchange routing information.
- The same commands repeated on Router2 (with its own local subnet, 192.168.2.0/24) let both routers advertise their networks to each other — once OSPF converges, each router automatically learns the route to the other's subnet without anyone manually typing static routes.